

11.2.18

AI25BTECH11012 - GARIGE UNNATHI

Question:

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of it.

Solution:

Let ABC be the triangle ,
E and F be the midpoints of AB and BC respectively .

$$\mathbf{E} = \frac{A + B}{2} \quad (0.1)$$

$$\mathbf{F} = \frac{B + C}{2} \quad (0.2)$$

The line joining these two points is

$$\mathbf{EF} = \frac{C - A}{2} \quad (0.3)$$

The line AC is

$$\mathbf{AC} = \mathbf{C} - \mathbf{A} \quad (0.4)$$

Hence ,

$$\mathbf{EF} = \frac{1}{2}\mathbf{AC} \quad (0.5)$$

Therefore the line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of it